

MAHMUDUL ALAM

Python Backend & Data Processing Engineer

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PROFESSIONAL SUMMARY

Backend software engineer with **3+ years of professional experience**, specializing in **Python** (Django and Django REST Framework) and currently contributing to production **FastAPI** services. Proven record of designing REST APIs, **Celery/Redis background-job systems**, and reliable **data pipelines** — scheduled payment reconciliation, LLM-based structuring of unstructured content, and idempotent ETL with atomic checkpointing. Builds **Excel/PDF reporting** and dashboard-backing APIs on **PostgreSQL** and MySQL. Two-time **ICPC Asia Dhaka Regional Divisional Champion**; works directly with international clients (EF SET C2 Proficient English).

TECHNICAL SKILLS

Languages: Python (type hints, OOP, exception handling, file processing), SQL, Rust, TypeScript/JavaScript

Backend: Django, Django REST Framework, FastAPI, Celery, REST API design, WebSockets (Django Channels)

Data & Storage: PostgreSQL, MySQL, Redis, Pandas, data ingestion & validation, CSV/JSON/plain-text parsing, Excel & PDF report generation (openpyxl, pdfkit), database indexing, transactions & query optimization

Auth & Security: JWT, OAuth2, WebAuthn/FIDO2, role-based access control, RSA request signing

DevOps & Testing: Docker, GitHub Actions (CI/CD), pytest, Locust, Selenium, Git, Linux, GCP (Google Cloud Platform), Nginx

PROFESSIONAL EXPERIENCE

Automation Solutionz — Software Engineer

Full-time, Remote | Jul 2025 – Present

- Maintain and extend the core **Django/DRF backend** of ZeuZ, an AI-powered enterprise test-automation platform, taking features from requirements through development, testing, and deployment.
- Contribute to **FastAPI** services that connect on-premises automation nodes to the platform server, including the ZeuZ AI backend.
- Process large **test-execution log datasets** with **Pandas** for the platform's reporting and data workflows.
- Built AI-powered Test Studio features that automate end-to-end test-case creation.
- Implemented a secure vault for sensitive test data with environment-variable-based secret injection.
- Added automation support for Linux desktop applications, extending platform compatibility to enterprise Linux environments.
- Act as a direct technical contact for the platform's international clients, from requirement clarification through delivery.

EJP — Full-Stack Developer

Part-time, On-site | Oct 2024 – Jul 2025

- Built an LLM **data-transformation pipeline** (Google Gemini, multi-API-key rotation) converting unstructured exam material into structured MCQ records, bulk-loaded into **PostgreSQL** via a batch **data-ingestion** endpoint.
- Designed the production **PostgreSQL** schema behind the platform's REST API (Rust/Actix-Web, served on a 1 vCPU / 1 GB VPS): handwritten **SQL migrations** covering foreign keys, triggers, and indexes, with order fulfillment wrapped in **atomic transactions**.
- Implemented generic paginated, filtered, and searchable REST endpoints backing admin dashboards for ~20 resource types, and integrated bKash and Nagad payment gateways, including RSA-signed (PKCS1v15) Nagad requests.

Begum Rokeya University, Rangpur — Backend Developer

Part-time, On-site | Aug 2024 – Oct 2024

- Developed a Next.js office file-tracking platform with **WebAuthn/FIDO2 passwordless authentication** for 1,000+ users, QR/barcode-based file identification, and automated email notifications.
- Designed **role-based access control** (NextAuth.js multi-provider) and an indexed MySQL schema (Prisma ORM) optimized for the hot access-key and barcode lookup paths.

- Architected a **Django REST Framework** backend from scratch to replace the company's legacy WordPress e-learning platform, substantially reducing hosting costs and increasing system capacity.
- Ran **Celery + Redis background jobs** with three Celery Beat schedules: 5-minute **payment reconciliation** re-verifying pending bKash/Nagad/DBBL transactions (auto-expiring after 30 minutes), 55-minute gateway token refresh, and 1-minute database health-check **monitoring** with automated recovery.
- Generated operational **reports** via Django admin actions: styled **Excel** admission/shipping workbooks (openpyxl) joining order, payment, and variant data, and **PDF** courier slips and packing summaries (pdfkit).
- Migrated three payment-gateway integrations from Java to Python with file-based **logging** across payment flows and post-payment fulfillment inside **atomic database transactions**.
- Wrote **pytest**-django API integration tests (registration, OTP auth, duplicate/invalid input flows) and **Locust** load-test scenarios; built filtered, searchable DRF endpoints (django-filter) backing dashboards.

FEATURED PROJECTS

Job-Application ETL Pipeline — Python, Selenium, cron

- Idempotent daily **data pipeline** that scrapes, deduplicates, validates, and LLM-ranks job postings: atomic checkpoint writes (tmp-file + os.replace); **resume-on-restart** that skips processed rows and retries errored ones with explicit error tracking; stable dedup keys; recovery of malformed LLM JSON output; and capped-backoff retries.

Live Meeting Transcriber — Python, ONNX, llama.cpp

- Offline Bengali/English meeting transcription with speaker diarization: **memory-bounded streaming** over unbounded audio (frame-level VAD segmentation, producer/consumer threads, byte-capped rolling context with eviction); cut WER from 42% to 28% via context replay with sequence-alignment dedup, and reduced inference latency 3.5×.

Network Tunneling Client — Rust, tokio, ~4.5k LOC

- SOCKS5 tunneling client with throughput-ranked upstream pooling, append-only connection **logs** with time-window retention and compaction, live traffic **monitoring**, and ~48 **unit and integration tests** (end-to-end protocol loopback, mocked HTTP auth) run in a pinned **Docker** test harness.

B12J Online Judge — Django, DRF, Django Channels, MongoDB

- Competitive-programming platform with real-time **WebSocket** verdict streaming (Daphne ASGI), djoser + Firebase OAuth2 authentication, and an isolated judge worker executing untrusted code under enforced time/resource limits with per-test-case verdict reporting.

Codeforces Tracker API — Django, DRF, PostgreSQL

- **External-API ETL** syncing full Codeforces submission history with verdict aggregation and **duplicate-record detection** and cleanup; PostgreSQL ArrayField/JSONField modeling behind DRF endpoints.

EDUCATION & ACHIEVEMENTS

- **B.Sc. in Computer Science and Engineering** — Begum Rokeya University, Rangpur (2020 – 2025).
- **ICPC Asia Dhaka Regional** — Divisional Champion (2022, 2024); three-time participant.
- **Codeforces Expert** (rating 1691) — codeforces.com/profile/mahbd.
- **EF SET English Certificate** — C2 Proficient (75/100).